

Osaid Khan

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Dear Hiring Team,

It is with great pleasure that I submit my candidacy for the **Backend Software Engineer** with **Pickle Robot Company**. I bring over three years of professional experience in full-stack web applications, distributed systems, and AI-driven automation, along with a strong academic foundation as a Master of Science in Computer Science graduate from Pace University, NY.

Through my experience across two engineering roles, I have built and delivered production-grade backend systems that directly reflect **the distributed system design and concurrency expertise this position demands**. As a Software Engineer at Sperse, I architected a concurrent multi-agent orchestration system using LangChain and LangGraph, enabling scalable, reliable, and performant tool execution across complex backend workflows. I also developed an MCP interaction layer that translates natural-language intents into safe, validated tool calls across 1,000+ platform endpoints — a task that required rigorous service design and a focus on high code quality to support automated actions. To keep these systems production-ready, I implemented a full observability stack using Langfuse and LangSmith, covering tracing and evaluation hooks to monitor system performance, reliability, and maintainability. Prior to Sperse, as a Software Engineer at Qualitest, I designed and developed a cloud-agnostic Storage Service using Python, PostgreSQL, and Redis that seamlessly handled real-world operational file operations across AWS S3 and Google Cloud Storage. I also developed event-driven backend workflows using AWS Step Functions and Lambda to handle long-running data jobs, ensuring system reliability within distributed architectures — both of which required **the kind of systems-level thinking and operational stability focus required for robotics infrastructure**. Together, these roles have prepared me to **design and build backend services supporting task planning and autonomy systems** and drive impact within an agile robotics environment.

My academic research experience has given me a rigorous foundation in model development, GPU-accelerated computing, and technical communication — skills that translate directly into **the engineering standards and collaborative culture Pickle Robot values**. As an AI Researcher at the Pace Artificial Intelligence Lab, I conducted applied research on StyleGAN and Stable Diffusion, leveraging CUDA, PyTorch, and LoRA on an HPC cluster where I managed complex concurrency and resource allocation. I was responsible for managing the full model development lifecycle — ensuring secure and efficient model serving within a distributed environment. Beyond building the systems myself, I led workshops on GPU programming and building AI Agents, where I contributed to the team's engineering standards by breaking down complex technical concepts for others. This experience has shaped me into an engineer who builds with rigor and communicates with clarity, directly mirroring **the technical discussions and knowledge-sharing central to this role**. I am eager to bring this discipline to **Pickle Robot Company and its mission to automate global supply chains**.

I have always been passionate about **prioritizing customer needs over technology** and “**being in the trenches**” to **solve real-world problems** and envision myself contributing my talent to the furthering of **Pickle's goal to automate inbound and outbound processes at 1 million warehouse doors**. I would love to discuss my interest in this position with your team and can be reached at +1 (945) 304-5781 or khanosaid726@gmail.com. Thank you for your time and consideration, I look forward to connecting!

Sincerely,
Osaid Khan